University of Moratuwa, Sri Lanka
Department of Electronic & Telecommunication Engineering

EN3160 – Fitting and Alignment

Assignment 02 | Dilhan W.A. | 230145E
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Repository: `https://github.com/ahesh-dilhan/en3160-assignment-02`

1. Sunflower detection using scale-space extrema

The supplied image is read with `cv.IMREAD_REDUCED_COLOR_4`, giving 360×360 pixels. The detector targets dark central disks rather than irregular petal outlines. For darkness $D = 1 - I_{\text{gray}}/255$, the response is $L(x, y, \sigma) = -\sigma^2 \nabla^2 (G_\sigma * D)$; positive maxima identify dark blobs in the original image. The ideal-disk radius estimate is $r = \sqrt{2}\sigma$.

```
L = np.stack([-s*s*gaussian_laplace(D, s) for s in sigmas])
peaks = (L == maximum_filter(L, size=(3, 3, 3))) & (L >= 0.12)
```

Parameters. There are 36 geometrically spaced scales, $\sigma \in [1.5, 16]$ pixels; end-scale maxima are excluded. The threshold is 0.12 and allowed disk overlap is 35% of the smaller area. A yellow-petal prior requires 50% yellow support in the annulus $[r, 2.2r]$, with angular coverage. A Hessian curvature ratio of at least 0.5 rejects elongated gaps. No Hough-circle detector is used.

(a) Reduced input: 360 x 360



(b) LoG detections; largest five numbered



Figure 1: Original-colour input and 53 detected candidates. The five largest central disks are numbered in the right panel.

Rank	x	y	σ (px)	r (px)	Response
1	282	338	7.106	10.050	0.351
2	106	255	5.801	8.204	0.365
3	179	327	5.801	8.204	0.346
4	178	261	5.422	7.668	0.353
5	274	258	4.736	6.698	0.347

Interpretation. The largest disk is centred at $(282, 338)$ with $r = 10.05$ pixels. Coordinates use the top-left origin of the reduced image; multiply positions and radii by four for original-image units. Colour and symmetry checks suppress foliage responses but miss distant, tilted or occluded flowers. These are candidate detections, not a verified flower count; the estimated radii describe central disks, not complete blossoms.

2. Custom RANSAC fitting: line followed by circle

Data and hypotheses. Listing 1 is reproduced with data seed 0: 50 circle points with centre $(2, 3)$, radius 10 and radial-noise standard deviation 0.625, plus 50 points from $y = -x + 2$ with vertical-noise standard deviation 1. Both RANSAC loops are coded directly. A line hypothesis uses two distinct points; a circle uses three non-collinear points. Degenerate samples are rejected. Each model uses 2,000 trials; the sequence starts with RNG seed 100.

Residuals and constraints. The line is $ax + by = d$ with $a^2 + b^2 = 1$ and $d \geq 0$. Its error is perpendicular distance, not vertical distance. The circle uses radial error. Both thresholds are 1.5, approximately 2.12 normal-noise standard deviations for the true line and 2.4 for the circle:

$$e_\ell(\mathbf{p}) = |ap_x + bp_y - d|, \qquad e_c(\mathbf{p}) = \big|\|\mathbf{p} - \mathbf{c}\|_2 - r\big|.$$

The first line needs at least 35 inliers. Its best consensus is removed, after which the circle needs at least 25. Total least squares refines the line; L-BFGS-B minimizes mean squared radial error with $r > 0$. Consensus membership is updated after refinement. Winning minimal samples and their unrefined models are retained separately.

```
line = ransac(X, "line", rng, threshold=1.5, minimum_consensus=35)
remaining = np.flatnonzero(~line.inliers)
circle = ransac(X[remaining], "circle", rng,
                threshold=1.5, minimum_consensus=25)
```

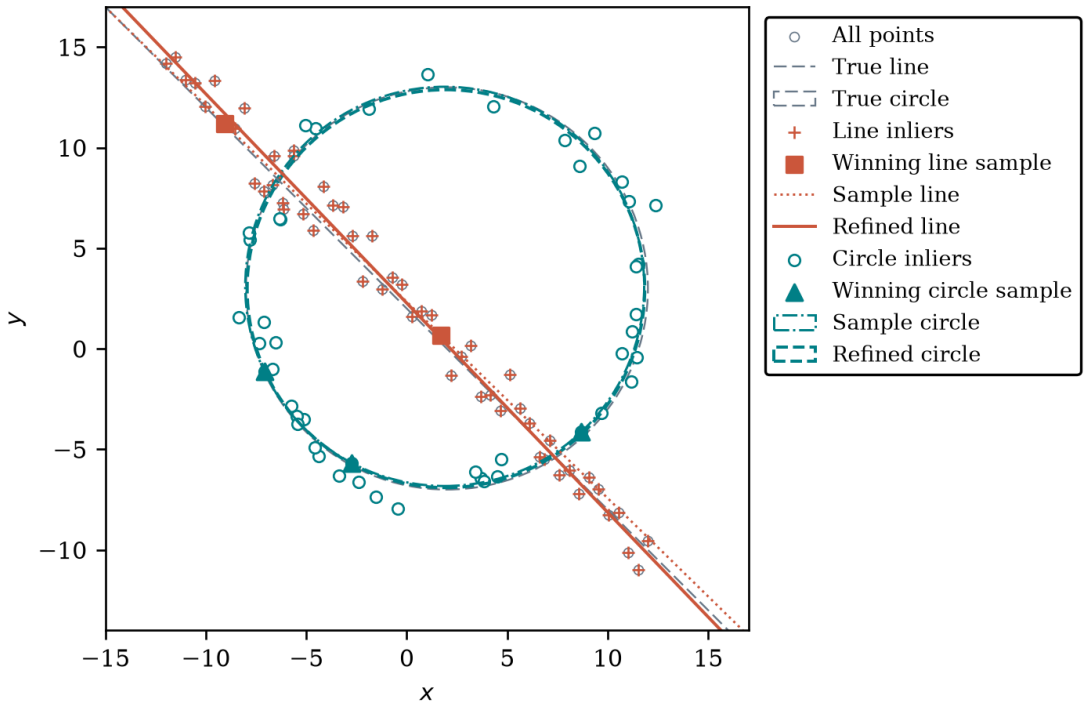


Figure 2: All points, winning line and circle samples, sample hypotheses, refined models and their consensus sets. Markers and line styles distinguish models while labels remain black.

Model	Refined parameters	Inliers	Inlier RMSE
Line	$a = 0.7210,\ b = 0.6930,\ d = 1.5724$	54	0.6706
Circle	$c_x = 1.9494,\ c_y = 2.9879,\ r = 9.8919$	46	0.5925

Iteration budget. For confidence $p = 0.99$ and a hypothetical inlier fraction $w = 0.5$, $N = \lceil \log(1-p)/\log(1-w^s) \rceil$ gives 17 trials for $s = 2$ and 35 for $s = 3$. The larger fixed budget accommodates contamination and lower inlier fractions after consensus removal; it does not guarantee success.

Circle first. Seed 0 also succeeds in this order: centre $(1.943, 3.008)$ and radius 9.930. Across data seeds 0–19, circle recovery occurs in 20/20 line-first runs and 19/20 circle-first runs, using centre and radius errors below 2 as the declared criterion. A very large circle can approximate a line and claim its points, so circle-first is vulnerable but does not always fail. Line-first also removes some genuine circle points near intersections, explaining its consensus of 54. Geometric refinement is local, not proof of a global optimum.

3. Planar homography and image compositing

Choice of image pairs. Two architectural photographs are paired with original graphic overlays: a fictional open-studio banner on a flat band of a modern facade, and a community-garden notice on a garage door. The first gives an oblique placement and the second a nearly frontal comparison. These notices do not describe actual events or imply endorsement by property owners.

Four manually selected destination corners are ordered top-left, top-right, bottom-right, bottom-left. They were selected from visual inspection; actual mouse clicks have not been recorded. The supplied `select_corners.py` script supports repeatable four-click selection. The saved coordinates and blend weights are listed below.



Figure 3: Two chosen architectural image pairs, retaining original colours. Each row compares the input photograph with its composited result.

Mapping and blending. The four source corners and four destination corners define H, mapping overlay coordinates to background coordinates: $\mathbf{x}' \sim H\mathbf{x}$. The graphic and an all-one coverage mask are warped with identical bilinear interpolation. Using premultiplied warped colour W and coverage M, the result is $C = B(1 - \alpha M) + \alpha W$. Black graphic pixels remain valid content; coverage is independent of pixel brightness.

```
H = cv.getPerspectiveTransform(source_corners, destination_corners)
W = cv.warpPerspective(overlay.astype(np.float32), H, size)
M = cv.warpPerspective(np.ones((h, w), np.float32), H, size)
result = background*(1-alpha*M[..., None]) + alpha*W
```

Pair	Destination corners (image pixels)	α
Studio	(215,405), (953,508), (953,556), (215,461)	0.82
Garden	(454,683), (782,683), (786,800), (450,800)	0.78

Assessment. Both homographies reproduce their selected corners to within 0.001 pixels. This tests point correspondence, not real-world accuracy: the same four pairs define the transform exactly and are not independent validation. Alpha blending retains some surface texture but does not model physical illumination, shadows, reflections or occlusion. The facade band is more perspective-distorted than the garage placement.

Image credits. Expect Best, *Modern Building Against Sky*, Pexels photo 323780; Binyamin Mellish, *Brown and Gray Painted House in Front of Road*, Pexels photo 1396122. Both were downloaded at width 1600 and modified with original graphics. Source pages: pexels.com, photo 323780; pexels.com, photo 1396122. License: `https://www.pexels.com/license/`. Full attribution is retained in the repository.

4. SIFT-based PCB registration and differences

Registration. The template is `AECC1.jpg`. SIFT extracts up to 8,000 features per image at maximum side 1600. `BFMatcher` uses L2 two-neighbour matching with a 0.72 ratio test. RANSAC estimates the target-to-template homography with a 3-pixel threshold at feature scale, confidence 0.999 and up to 10,000 trials. The mapping is converted to original-image coordinates before warping all five targets.

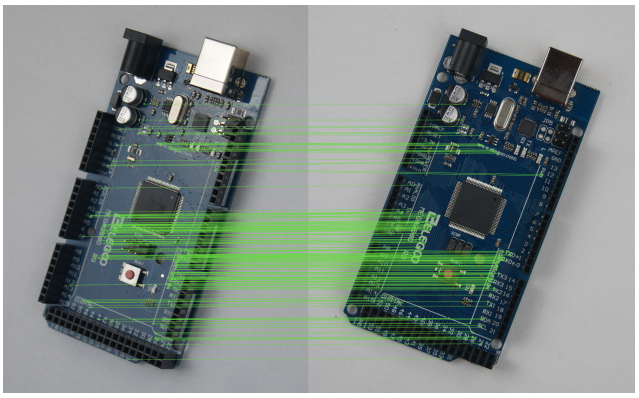


Figure 4: ADAD1 to AECC1: green lines show accepted RANSAC matches.

Registration confidence. At least eight geometric inliers are required. AAAA1 has only 15, with an inlier hull spanning 10.1% of the full template image, so its large viewpoint change makes it less reliable than AECC2–4. Full-size match visualizations and alignment overlays are saved in the experiment output folders.

Valid overlap. An all-one image is warped with the same interpolation. Partly padded pixels and a one-pixel boundary margin are excluded. This prevents black warp borders from inflating the difference statistics.

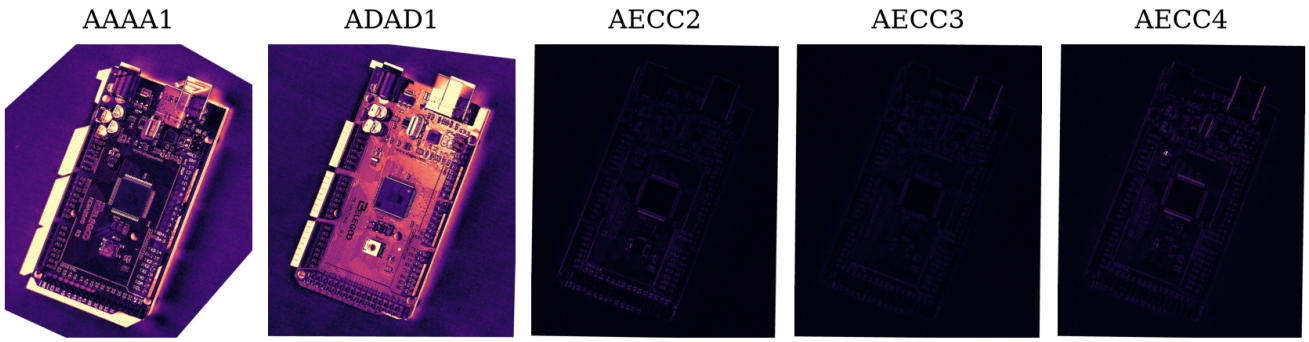


Figure 5: Mean absolute RGB differences, displayed with the original magma colour map over a common 0–100 range: dark is zero and pale yellow is 100 or above. Blank white margins outside the warped footprint are invalid overlap. Numerical metrics use the full raw values.

Target	Matches	Inliers	Median error*	Overlap	Raw MAE
AAAA1	38	15	0.944	89.0%	29.31
ADAD1	252	187	0.439	98.1%	30.62
AECC2	3314	3068	0.210	98.3%	3.52
AECC3	3978	3664	0.167	98.8%	3.29
AECC4	3171	2913	0.231	98.4%	4.07

*Median inlier reprojection error, in feature-image pixels. MAE averages valid RGB-channel differences over the entire overlap, including background. It is not a board-only defect score. No exposure normalization is applied.

```
H_full = np.linalg.inv(template_scale) @ H_small @ source_scale
aligned = cv.warpPerspective(target, H_full, template_size)
difference = cv.absdiff(aligned, template)
difference[~valid] = 0
```

Why raised components differ. The USB Type B port, capacitors and reset button lie above the PCB plane. Viewpoint changes cause parallax and occlusion that one planar homography cannot remove. Specular reflections, shadows, exposure, residual misalignment and possible component differences also contribute. AAAA1 and ADAD1 have much larger raw MAE than AECC2–4, consistent with their stronger viewpoint and appearance changes. Bright residuals alone do not establish a manufacturing defect.

Reproducibility and references. Ten geometry tests pass, including unit-normal fitting, robust homography recovery, black-overlay preservation and valid-overlap masking. Assignment: R. Rodrigo, *EN3160 Assignment 2 on Fitting and Alignment*, 9 October 2026. Optimization background: S. Boyd and L. Vandenberghe, *Convex Optimization*, §1.1, Stanford; SciPy minimize documentation. Code, parameters and genuine commit history: `https://github.com/ahesh-dilhan/en3160-assignment-02`.